From Node To Range

The first four levels (Node, Channel, Database, Range) are specifically designed to explore the rules. I tried a special design approach, hoping that players could make inferences through their attempts.

If we fail to figure out some of the rules in the first three levels, we are very likely to hit the wall in the fourth level (Range).

Let me explain:

Warning: The solutions are revealed on the following pages, and we can stop anytime and play through the levels at wer own pace.

About 'Line': All 'line' below are actually line segments , which has two endpoints and finite length.

Game Control

This demo might feel unplayable at first because there are no on-screen button hints. Although the level design itself uses a guided approach, misunderstandings are still possible. Therefore, here is a clear explanation of the controls:

The main game interface consists of two parts: the **top toolbar** and the **level canvas area** below.

Top Toolbar (from left to right):

- **"UNDO"** – Reverses wer last action.
- **"REDO"** – Restores an action we just undid.
- **"Previous Level"** (arrow left) – Goes to the previous level.
- **Level indicator** – Shows "Current Level / Total Levels" (e.g., 2/8).
- **"Next Level"** (arrow right) – Goes to the next level.
- **"RESET"** – Clears all player-drawn lines and resets the current level.
- **"THEME"** – Switches the UI color scheme.
- **"EDIT"** – Opens the level editor. *(Note: It is recommended not to enable this before completing the levels, as it may spoil the solution.)*

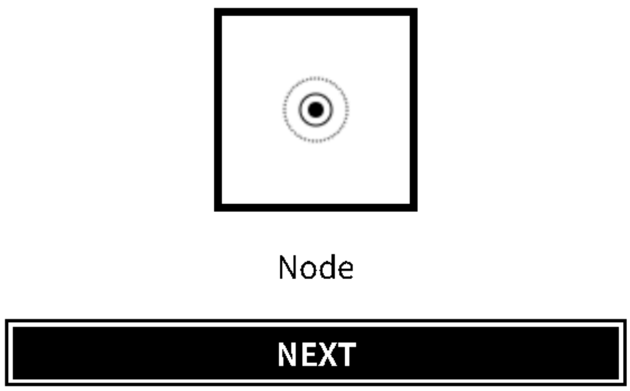
Level Canvas Area:

- This area consists of interactive points(circles) that players can click.
- **Right-clicking** on the canvas has the same effect as pressing **"UNDO"** – it removes wer last drawing action.
- Below the canvas is the **level progress bar**.
- When all rules are satisfied, the progress bar fills up completely and displays **"NEXT"**.
- This means the level is complete. Click **"NEXT"** to advance to the next level.

Level 1 - Node

We click the only point to fulfill the level, and we will also be able to know:

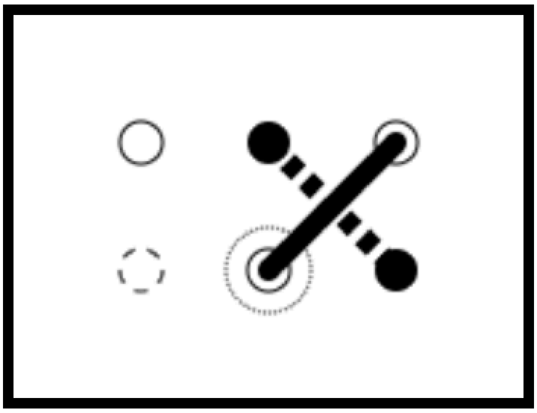
- The solid-line circles can be occupied, and we need to occupy at least one point on the canvas in order to pass the level.
- Furthermore, now we all know what the occupied points look like.
- When we complete the level, the progress bar below the canvas will be filled up, and "NEXT" will appear (and we can click to move on to the next level)



Level 2 - Channel

Level 2 might require several attempts, but most of the time we will eventually find the correct answer:

- When we click multiple points in sequence, a straight line will connect them one after another, forming a polyline through our clicks.
- When our line crosses the dashed line, the level is completed.
- Dashed circles and fully filled solid circles (the endpoints of the dashed line) cannot be occupied.
- The starting and ending points of the answer can be swapped (the connection has no direction).

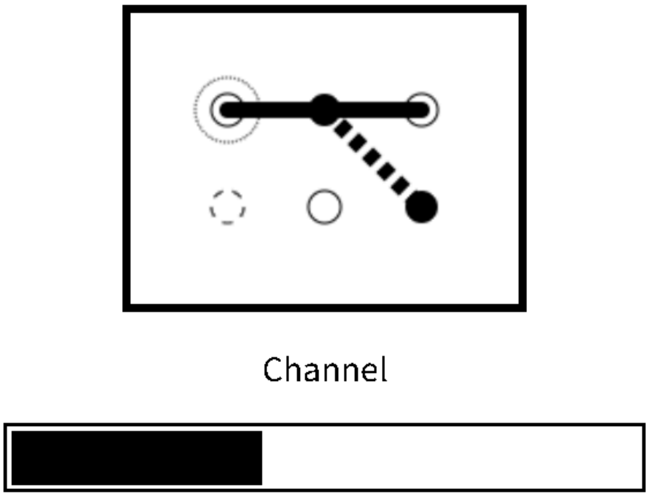


Channel

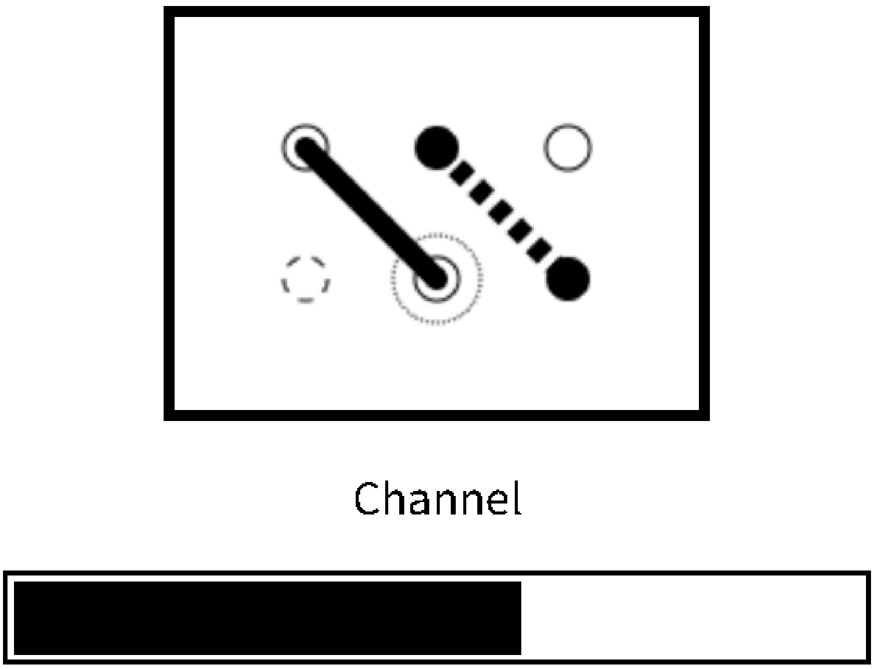
NEXT

However, in this level, failed attempts also carry information:

- Crossing an endpoint of a dashed line does not count as crossing a dashed line.



- Our line seems to have to cross the dashed line.



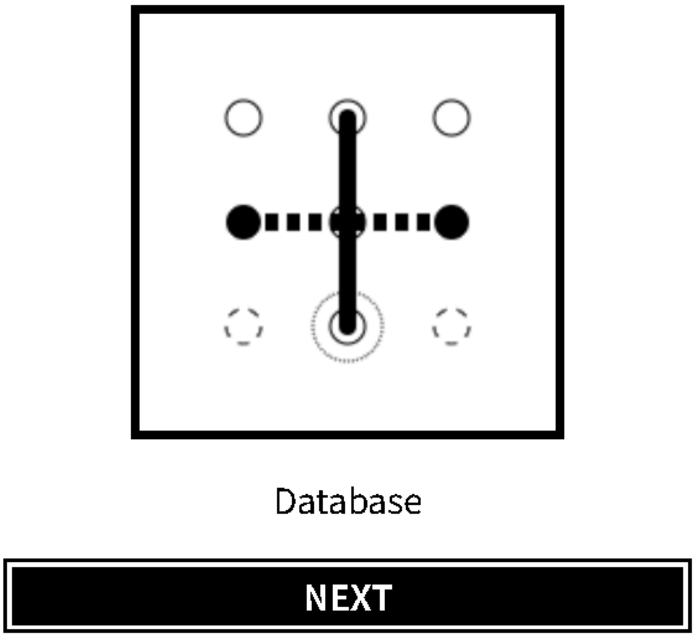
- It seems that extra lines are not allowed or that extra lines violate some rule.

Channel

Level 3 - Database

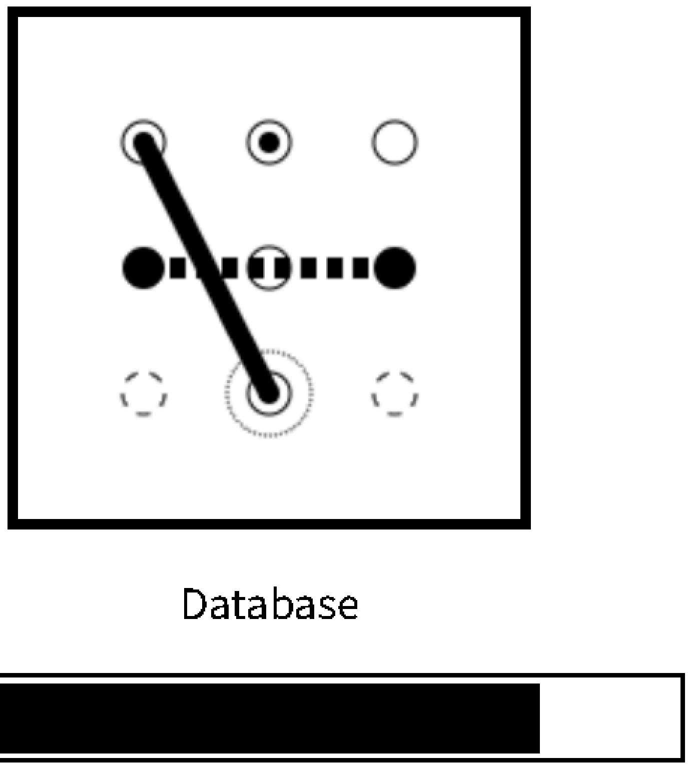
In the third level, we can see that there is already a point on the canvas that seems to be occupied. If we start from the occupied point, after several attempts we can obtain the answer:

- The points that have already been occupied are the endpoints of our line.
- The order of the connection lines doesn't seem to matter much (both from the occupied point at the top to the point at the bottom and from the point at the bottom to the occupied point at the top can pass).

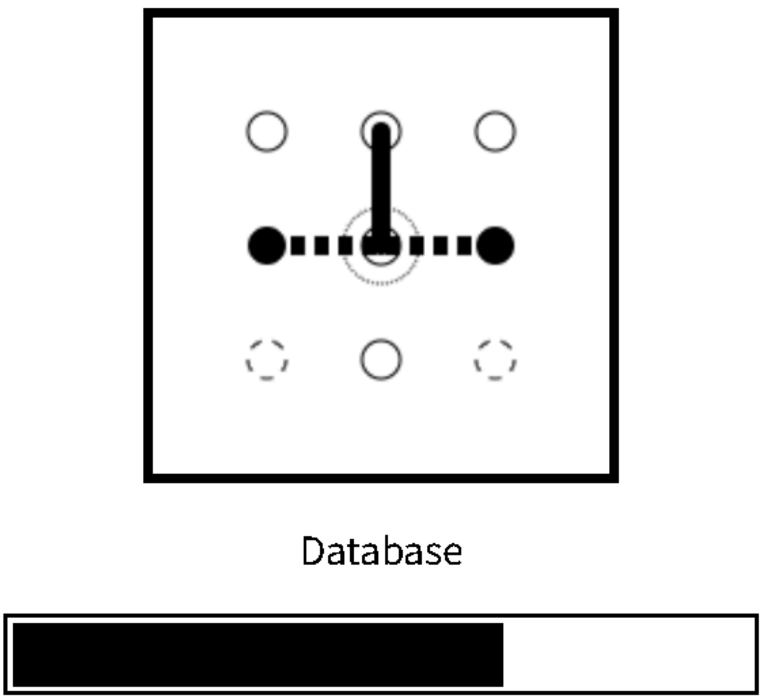


In trying, we may also find:

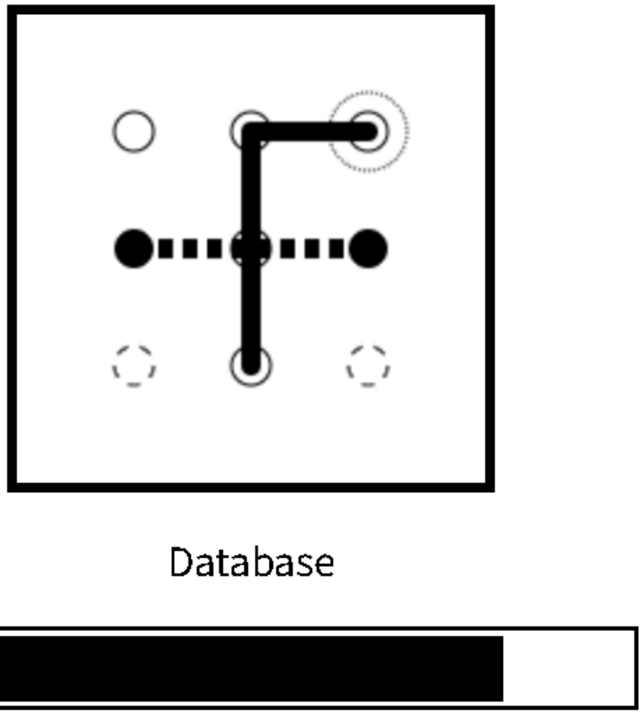
- The points that have already been occupied must become one of the endpoints of our lines.



- The dashed line passing through our endpoints does not count as our intersection with the dashed line.



- Additional lines will disrupt the completion of the level progress.

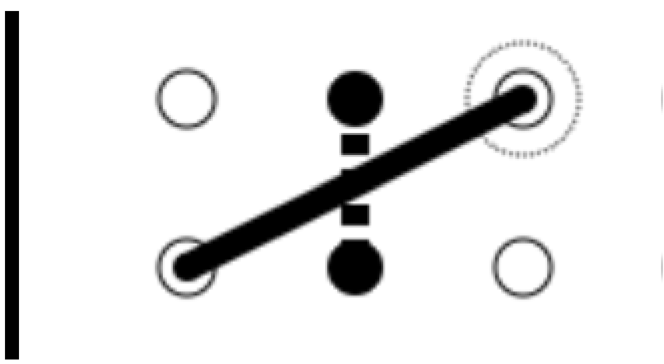


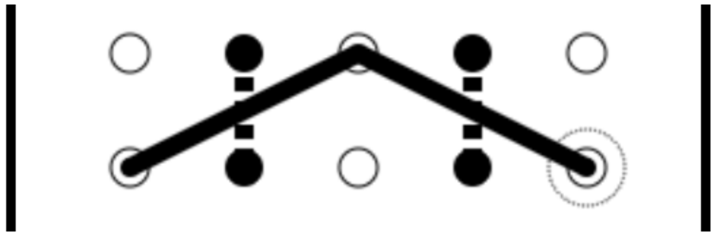
Level 4 - Range

The fourth level might be a bit complicated because it involves the interrelationships of multiple lines. This time, we will start our reasoning:

- We can summarize the existing rules:
    - The points we occupy will sequentially form a polyline.
    - The line we construct must cross the existing dashed line.
    - The endpoints of the dashed lines and dashed circles cannot be occupied.
    - Any additional lines that do not meet the rules in addition to the required ones will disrupt the progress of the level.

Regarding the crossing rules, for the case of a dashed line of length 1, there is basically only one solution (since intersections at the endpoints do not count):





Then, we suddenly found that it only took one step to complete the level (which won't be shown here).

But in this level, some other rules can still be tried out:

- An already occupied point needs only to be used in the polyline; it does not need to be a start or end point.
- Crossing the same dashed line twice will set the level progress back.
- Our polyline cannot self-intersect.
- Additional lines apart from the crossing line will set the level progress back.

The Rules

[ROT13] Pyvpx ba gur cbvagf gb qenj n cbylyvar. Rnpu yvar (frtzrag) zhfg pebff bar qnfurq yvar. Rnpu qnfurq yvar pna or pebffrq ol rknpgyl bar yvar (frtzrag), naq rnpu yvar (frtzrag) pna pebff rknpgyl bar qnfurq yvar.